Linear Algebra and Its Applications

Section 8.5: Polytopes

Detailed worked solutions

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Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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8.5 Polytopes

A polytope is the convex hull of finitely many points. Its minimal representation consists exactly of its extreme points, or vertices. For a linear functional, the value at a convex combination is the same combination of the values at its generators. Hence its maximum is the largest generator value; the full maximizing set is the convex hull of *all* generators tying for that largest value. The analogous statement holds for minima. A k-face has dimension k; vertices have dimension zero, edges dimension one, and facets of an n-polytope dimension $n-1$.

Practice problem 8.5.P1: Turn five inequalities into a vertex list

In the first quadrant the constraints are $x+3y\le 12$, $x+2y\le 9$, $2x+y\le 12$. The axis vertices are $(0,0),(6,0),(0,4)$, obtained by imposing the most restrictive intercept on each axis. Pairwise boundary intersections are $(3,3),(5,2),(24/5,12/5)$. The last violates $x+2y\le 9$, because $24/5+24/5=48/5>9$. The first two pass the remaining inequalities. Thus the minimal representation is

$$P=\operatorname{conv}\{(0,0),(6,0),(5,2),(3,3),(0,4)\}.$$

Each retained point is a genuine corner of the feasible polygon; no retained point is a convex combination of the others. The printed final sentence calls these "three" vertices while listing five; there are five.

Exercise 8.5.1: All maximizers of three functionals on a triangle

At $p_1=(1,0), p_2=(2,3), p_3=(-1,2)$, the functional values are

	p_1	p_2	p_3	max	all maximizing points
$x-y$	1	-1	-3	1	p_1
$x+y$	1	5	1	5	p_2
$-3x+y$	-3	-3	5	5	p_3

These respectively answer (a), (b), (c). To justify completeness, write $x=\sum_i \lambda_i p_i$ with nonnegative coefficients of sum one. If a positive coefficient were attached to a vertex with value below the maximum, their weighted average would be strictly below that maximum. Here each maximum occurs at just one vertex, so no other point can attain it.

Exercise 8.5.2: Two maximizing edges

For $p_1=(0,-1), p_2=(2,1), p_3=(1,2)$, evaluation gives

	p_1	p_2	p_3	max	all maximizing points
$x+y$	-1	3	3	3	$\operatorname{conv}\{p_2,p_3\}$
$x-y$	1	1	-1	1	$\operatorname{conv}\{p_1,p_2\}$
$-2x+y$	-1	-3	0	0	p_3

In each tied case every point on the listed edge has the same value by linearity. A convex combination involving the lower-value vertex with positive weight has smaller value, so the whole maximizing set is exactly that edge.

Exercise 8.5.3: Minima for the first triangle

Use the values from Exercise 8.5.1 and select the smallest in each row. For (a), the minimum of $x - y$ is -3, attained only at p_3. For (b), the minimum of $x + y$ is 1, attained at every point of the edge $\text{conv}\{p_1, p_3\}$ and nowhere else. For (c), the minimum of $-3x + y$ is -3, attained exactly on $\text{conv}\{p_1, p_2\}$. The weighted-average argument used for maxima also proves these sets contain all minimizers.

Exercise 8.5.4: Minima for the second triangle

From the values in Exercise 8.5.2: (a) $x + y$ has minimum -1 only at p_1; (b) $x - y$ has minimum -1 only at p_3; (c) $-2x + y$ has minimum -3 only at p_2. No other point can attain one of these strict vertex minima, because a combination using another vertex would increase the weighted average.

Exercise 8.5.5: A polygon bounded by two sloping lines

The inequalities are $x + 2y \leq 10$, $3x + y \leq 15$, $x, y \geq 0$. The axis endpoints are $(5, 0)$ and $(0, 5)$. Solve both boundary equations: substitute $y = 15 - 3x$ into the first, obtaining $x + 30 - 6x = 10$, so $x = 4, y = 3$. This intersection is in the first quadrant. Together with the origin these are all corners:

$$P = \text{conv}\{(0, 0), (5, 0), (4, 3), (0, 5)\}.$$

Each corner is the meeting point of two boundary lines of the feasible polygon, and each lies on its boundary with a change in direction, so none is redundant in the minimal representation.

Exercise 8.5.6: Another two-constraint polygon

Here $2x + 3y \leq 18$, $4x + y \leq 16$, $x, y \geq 0$. The axis endpoints are $(4, 0), (0, 6)$. The two sloping boundaries intersect where $y = 16 - 4x$ and $2x + 3(16 - 4x) = 18$, giving $x = 3, y = 4$. Thus

$$P = \text{conv}\{(0, 0), (4, 0), (3, 4), (0, 6)\}.$$

All listed points satisfy all constraints and are the successive corners of the feasible polygon, hence give its minimal representation.

Exercise 8.5.7: Three boundaries meet at one corner

The constraints are $x + 3y \leq 18$, $x + y \leq 10$, $4x + y \leq 28$, with nonnegative coordinates. The restrictive axis intercepts are $(7, 0)$ and $(0, 6)$. Solving any pair of the sloping equalities gives the same point $(6, 4)$, which satisfies all three with equality. Hence there is only one non-axis corner, not three:

$$P = \text{conv}\{(0, 0), (7, 0), (6, 4), (0, 6)\}.$$

The inequality $x + y \leq 10$ touches this corner but introduces no new vertex.

Exercise 8.5.8: A redundant inequality

The constraints are $2x + y \leq 8$, $x + y \leq 6$, $x + 2y \leq 7$, with $x, y \geq 0$. Their restrictive axis endpoints are $(4, 0), (0, 7/2)$. The first and third boundaries meet at $(3, 2)$, which satisfies $x + y \leq 6$. The intersections with the middle boundary are $(2, 4)$ and $(5, 1)$; they violate respectively $x + 2y \leq 7$ and $2x + y \leq 8$ and must be discarded. Therefore

$$P = \text{conv}\{(0, 0), (4, 0), (3, 2), (0, 7/2)\}.$$

The discarded points are intersections of boundary lines, but they are not feasible and therefore cannot be vertices of this polytope.

Exercise 8.5.9: An extreme point that is not an exposed vertex

Let $D = \{(x, y) : x^2 + (y - 1)^2 \leq 1\}$ and $K = \text{conv}(D \cup \{(3, 0)\})$. Every point of K has $y \geq 0$, and its points with $y = 0$ form precisely the segment $[(0, 0), (3, 0)]$. Indeed the disk itself touches $y = 0$ only at the origin. If the origin were a strict convex combination of two distinct points in K, their nonnegative heights would both have to be zero. Their nonnegative horizontal coordinates on that segment could then average to zero only if both points were the origin. Thus the origin is extreme. However, the only supporting line to the disk at the origin is its tangent $y = 0$. Any supporting line to K there must also support the contained disk, so it must be that same line. It intersects K in the whole segment to $(3, 0)$, not in a singleton.

Conclusion. The origin is extreme but is not a vertex in the section's supporting-hyperplane (zero-dimensional exposed-face) sense.

Exercise 8.5.10: A closed unbounded counterexample to profile generation

Take the ray $S = \{(x, 0) : x \geq 0\}$. It is closed and convex. Its origin is extreme, while every point with $x > 0$ is the midpoint of two other ray points. Thus its profile is $\{(0, 0)\}$. The convex hull of that profile is just the singleton, not the whole ray. This shows why closedness without boundedness is insufficient.

Exercise 8.5.11: A bounded but nonclosed counterexample

Take the half-open segment $S = \{(x, 0) : 0 \leq x < 1\}$. It is bounded and convex. Its profile is again just the origin, since any other point has two distinct nearby points on either side inside the segment. The hull of the nonempty profile is a singleton, not S. This time the missing hypothesis is closedness.

Exercise 8.5.12: Face counts for simplices

(a) Count the faces of the five-simplex.

A five-simplex has six vertices; each k-face is determined by choosing $k + 1$ of them. Thus its counts for $k = 0, 1, 2, 3, 4$ are $6, 15, 20, 15, 6$. Their alternating sum is $6 - 15 + 20 - 15 + 6 = 2$, equal to $1 + (-1)^4$, as Euler's formula requires.

(b) Display the pattern and explain it.

$n\backslash k$	0	1	2	3	4
1	2	–	–	–	–
2	3	3	–	–	–
3	4	6	4	–	–
4	5	10	10	5	–
5	6	15	20	15	6

The general formula is $f_k(S^n) = \binom{n+1}{k+1}$ for $0 \le k \le n-1$. Affine independence ensures that each chosen $k+1$ vertices defines a k-simplex face and that different subsets give different faces. Dashes denote dimensions not belonging to the list of proper faces.

Exercise 8.5.13: Face counts for hypercubes

(a) Count a five-cube and check Euler.

Choose k coordinates allowed to vary and fix each remaining coordinate at either endpoint. Thus $f_k(C^n) = 2^{n-k}\binom{n}{k}$. For $n = 5$ the counts are $32, 80, 80, 40, 10$; $32 - 80 + 80 - 40 + 10 = 2 = 1 + (-1)^4$.

(b) The requested chart.

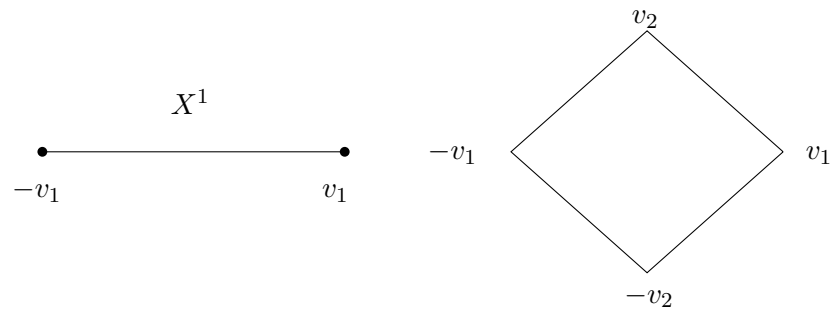
$n\backslash k$	0	1	2	3	4
1	2	–	–	–	–
2	4	4	–	–	–
3	8	12	6	–	–
4	16	32	24	8	–
5	32	80	80	40	10

The coordinate-choice argument proves the formula, rather than only predicting it from these few values.

Exercise 8.5.14: Crosspolytopes in increasing dimensions

(a) Draw the first two cases.

X^1 is the segment joining $-v_1$ to v_1. X^2 is the parallelogram with successive vertices $v_1, v_2, -v_1, -v_2$. Below are representatives using standard-coordinate generators; an invertible linear change gives the analogous pictures for any independent generators.



(b) The three-dimensional case.

There are 6 vertices, 12 edges, and 8 triangular facets. The standard-coordinate version is a regular octahedron; for arbitrary independent generators it is a linearly distorted octahedron with the same face counts.

(c) The four-dimensional case.

Its counts are $8, 24, 32, 16$. The alternating sum is $8 - 24 + 32 - 16 = 0 = 1 + (-1)^3$, as required.

(d) Explain the general formula.

A proper face selects $k+1$ of the n opposite vertex pairs, and selects one sign from each selected pair; both opposite vertices cannot belong to one proper face. These choices give

$$f_k(X^n) = 2^{k+1}\binom{n}{k+1}, \qquad 0 \le k \le n-1.$$

Every such selection is a simplex face: for the standard crosspolytope a supporting functional assigning the selected signed coordinates value one and all other coordinate coefficients zero exposes exactly that face. Linear equivalence preserves the face structure for general independent generators.

Exercise 8.5.15: Face counts for a pyramid

Let the $(n-1)$-dimensional base be Q and the apex lie outside its affine hull.

(a) Vertices.

All base vertices remain vertices and the apex adds one, so $f_0(P^n) = f_0(Q) + 1$.

(b) Intermediate dimensions.

A k-face not containing the apex is a k-face of Q. A k-face containing the apex is the pyramid over a $(k-1)$-face of Q. Thus $f_k(P^n) = f_k(Q) + f_{k-1}(Q)$ for $1 \le k \le n-2$.

(c) Facets.

There is one base facet Q; each of its $(n-2)$-faces joined to the apex gives a side facet. Therefore $f_{n-1}(P^n) = 1 + f_{n-2}(Q)$, for $n \ge 2$. These types are disjoint and cover every face of a pyramid.

Exercise 8.5.16: Four statements about profiles and simplices

(a) **True.** This is the definition of a polytope.

(b) **True.** An extreme point cannot lie strictly between two other points of the convex set. If $p \in uv$ and $p \neq u$, it must equal v.

(c) **False.** Compactness is necessary in the profile theorem; the ray in Exercise 8.5.10 is a counterexample.

(d) **True.** A four-simplex has five vertices; omit one at a time to obtain its five facets, each a three-simplex, or tetrahedron.

Exercise 8.5.17: Four more statements about faces

(a) **False.** A three-dimensional cube has six square facets.

(b) **True.** For polytopes the minimal-representation theorem identifies extreme points with vertices. The distinction seen for a nonpolytope in Exercise 8.5.9 does not occur for polytopes.

(c) **False.** A maximizing point need not itself be extreme. On the square $[-1,1]^2$, $f(x,y)=x$ is maximal at $(1,0)$, the midpoint of the edge joining $(1,-1)$ and $(1,1)$.

(d) **True.** A two-dimensional polytope is a polygon; each edge joins successive vertices around its boundary, giving equally many vertices and edges.

Exercise 8.5.18: Removing an extreme point preserves convexity

Suppose v is extreme and $x,y \in S \setminus \{v\}$. Their segment lies in S by convexity and cannot pass through v in its interior, by extremality; neither endpoint is v. Thus the whole segment lies in $S \setminus \{v\}$, proving that set convex. Conversely, if $S \setminus \{v\}$ is convex and v were strictly between some distinct $x,y \in S$, neither could equal v. Their segment would lie in the set with v removed, contradicting its containing v. Hence v is extreme. Singleton cases work because the empty set is convex.

Exercise 8.5.19: Add two positive dilations of one convex set

An arbitrary element of $cS+dS$ is $cx+dy$ with $x,y \in S$. Since $c,d>0$,

$$cx+dy=(c+d)\left(\frac{c}{c+d}x+\frac{d}{c+d}y\right)\in(c+d)S$$

by convexity. This proves one inclusion. Conversely, for each $s \in S$, $(c+d)s=cs+ds \in cS+dS$, proving the reverse inclusion. Thus $cS+dS=(c+d)S$.

Exercise 8.5.20: Convexity cannot be omitted

Take $S=\{(0,0),(1,0)\}$ and $c=d=1$. Then $S+S=\{(0,0),(1,0),(2,0)\}$, whereas $2S=\{(0,0),(2,0)\}$. The point $(1,0)$ occurs only in the sum. Thus the equality from Exercise 8.5.19 can fail for a nonconvex set.

Exercise 8.5.21: A Minkowski sum of convex sets is convex

Take two points $x=a_1+b_1$ and $y=a_2+b_2$ in $A+B$. For $0 \le t \le 1$,

$$(1-t)x+ty=((1-t)a_1+ta_2)+((1-t)b_1+tb_2).$$

The first parenthesis belongs to A and the second to B by their convexity. Thus their sum is in $A+B$. This proves the entire segment is included, as required.

Exercise 8.5.22: Why the list of regular polyhedra has five entries

(a) Count incidences in two ways.

Every one of the r facets has k edges, giving kr facet-edge incidences. Each edge belongs to two facets, giving $2e$; hence $kr = 2e$. Likewise every one of the v vertices meets s edges, while every edge has two endpoints, so $sv = 2e$.

(b) Substitute those counts into Euler's formula.

From $v - e + r = 2$, substitute $v = 2e/s$ and $r = 2e/k$, then divide by $2e$:

$$\frac{1}{s} + \frac{1}{k} = \frac{1}{2} + \frac{1}{e}.$$

(c) Exhaust the possible integer pairs.

A face has at least three sides, so $k \geq 3$; at least three edges meet at any vertex, so $s \geq 3$. The equation makes $1/s + 1/k > 1/2$. If $k \geq 6$, its left side is at most $1/3 + 1/6 = 1/2$, impossible. For $k = 3$, the strict inequality gives $s < 6$, so $s = 3, 4, 5$. For $k = 4$, it gives $s < 4$, so $s = 3$. For $k = 5$, it gives $s < 10/3$, again $s = 3$. The counts follow from $e = (1/s + 1/k - 1/2)^{-1}$ and the incidence formulas:

k	s	v	e	r	polyhedron
3	3	4	6	4	tetrahedron
4	3	8	12	6	cube
3	4	6	12	8	octahedron
5	3	20	30	12	dodecahedron
3	5	12	30	20	icosahedron

These are the five realized regular convex polyhedra named in the exercise, and the integer argument rules out every other possibility.